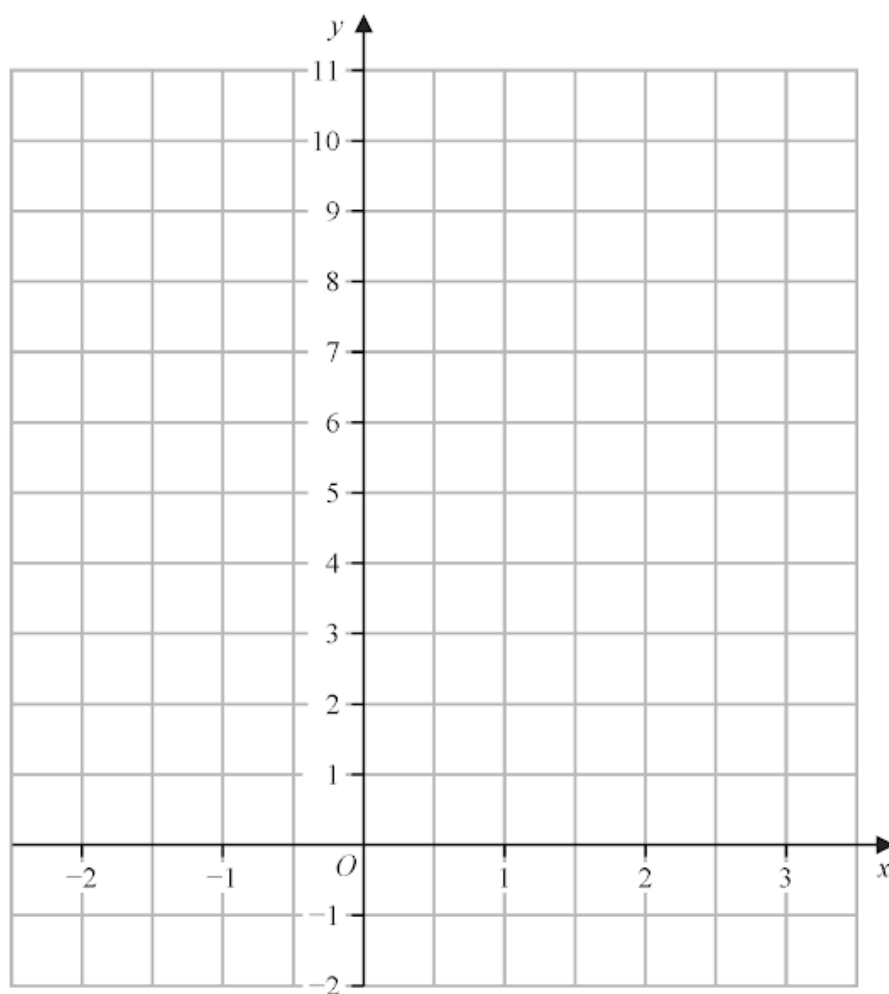


4MA1_2F_ch3_sequences_functions_graphs

Nguyen Khac Kiem PhD

1. 4MA1/2FR_Summer_2024_Q13

On the grid, draw the graph of $y = 5 - 2x$ for values of x from -2 to 3



(3)

2. 4MA1/2FR_Summer_2024_Q7

Here are the first five terms of a number sequence.

-3 5 13 21 29

(a) (i) Write down the next term of the sequence.

(1)

(ii) Explain how you worked out your answer.

(1)

(b) Explain why 326 cannot be a term of the sequence.

(1)

3. 4MA1/2F_Summer_2024_Q18

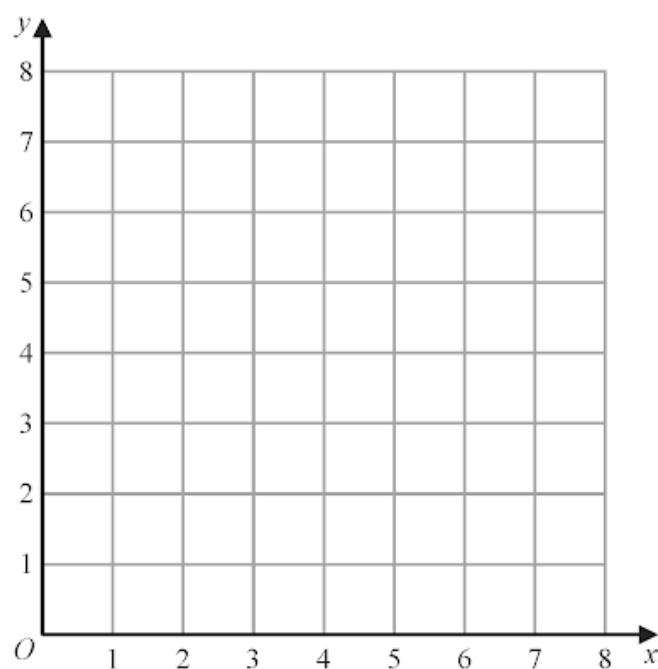
(a) On the grid, draw the straight line with equation

(i) $y = 2$

(ii) $x = 6$

(iii) $y = x + 1$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$y \geq 2$$

$$x \leq 6$$

$$y \leq x + 1$$

Label the region **R**

(1)

4. 4MA1/2F_Summer_2024_Q4

Here are the first five terms of a number sequence.

106 99 92 85 78

(a) (i) Write down the next term of the sequence.

(1)

(ii) Explain how you found your answer to part (a)(i)

(1)

The 9th term of the sequence is 50

(b) Work out the 12th term of the sequence.

(1)

Umberto says 7 is a term in the sequence.

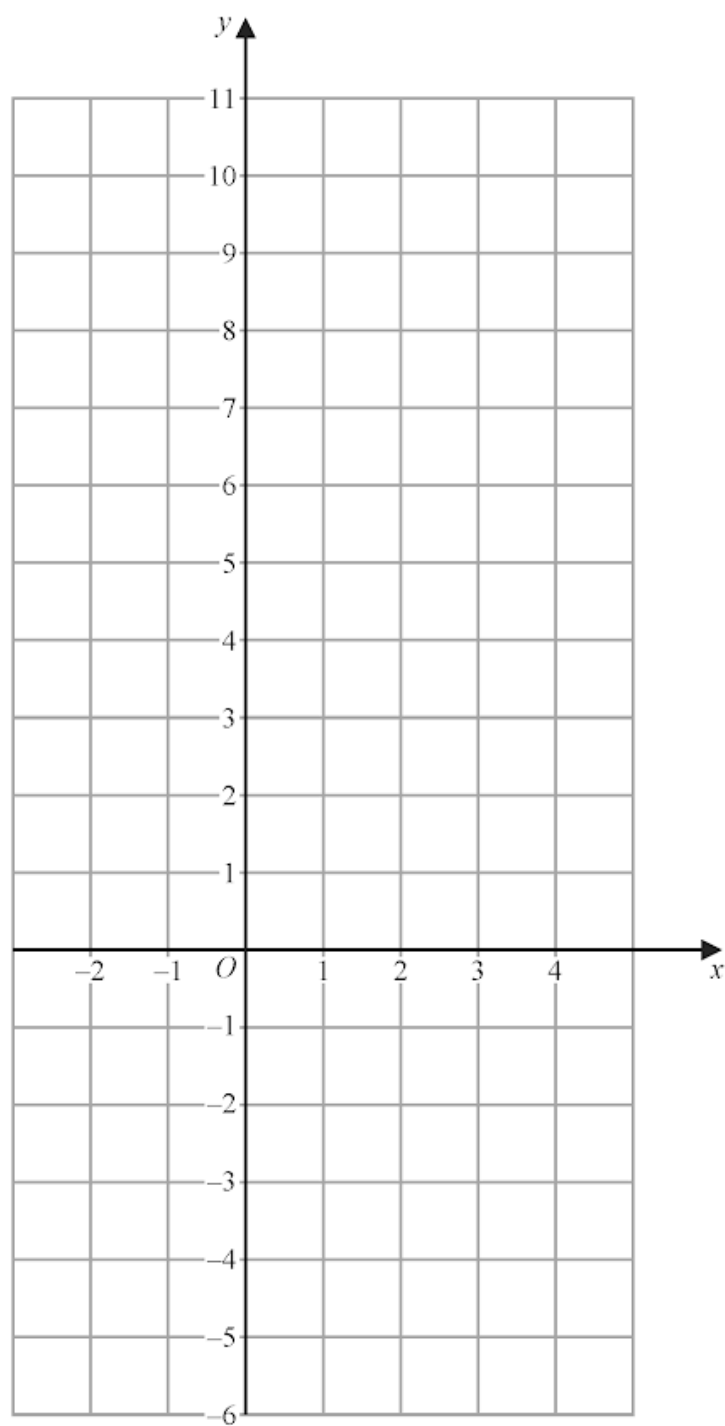
Umberto is wrong.

(c) Explain why.

(1)

5. 4MA1/2FR_Winter_2023_Q19

On the grid, draw the graph of $5x + 2y = 10$ for values of x from -2 to 4



3 marks

6. 4MA1/2FR_Winter_2023_Q6

Here are the first five terms of a number sequence.

4 10 16 22 28

(a) (i) Write down the next term of the sequence.

.....
(1)

(ii) Explain how you worked out your answer.

.....
(1)

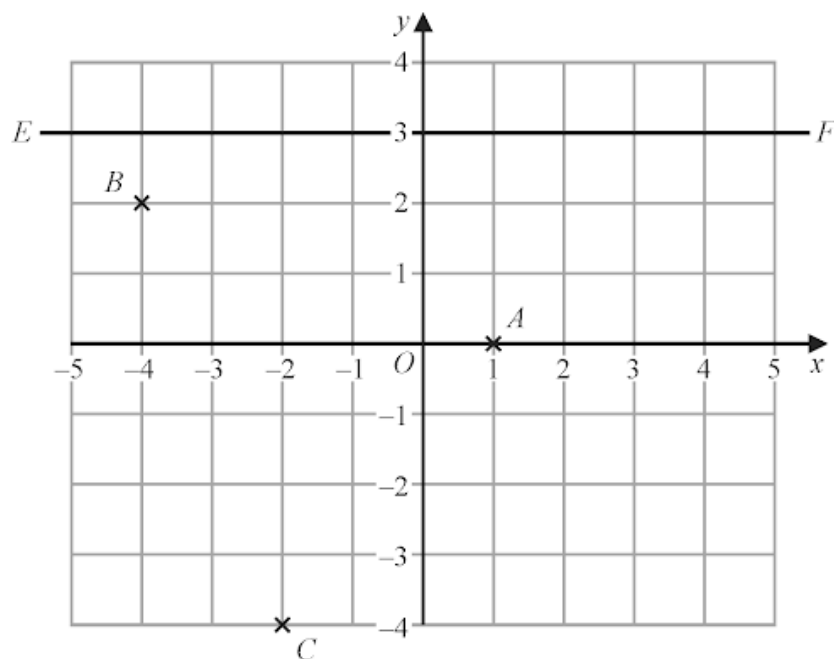
(b) Work out the 13th term of the sequence.

.....
(1)

(c) Explain why 467 cannot be a number in the sequence.

.....
.....
(1)
.....
.....
.....
.....
.....

The diagram shows three points, A , B and C , and a line EF on a grid.



(a) Write down the coordinates of the point A

(..... ,)
(1)

The coordinates of the point D are $(3, -2)$

(b) On the grid, mark with a cross ($\times$) the position of D
Label the cross D

(1)

(c) Find the coordinates of the midpoint of BC

(..... ,)
(2)

(d) Write down the equation of the line EF

.....
(1)

8. 4MA1/2F_Winter_2023_Q21

The points A and B are on a coordinate grid.

The coordinates of A are $(6, 4)$

The coordinates of B are $(17, j)$ where j is a constant.

The midpoint of AB has coordinates $(k, 15)$ where k is a constant.

Find the value of j and the value of k

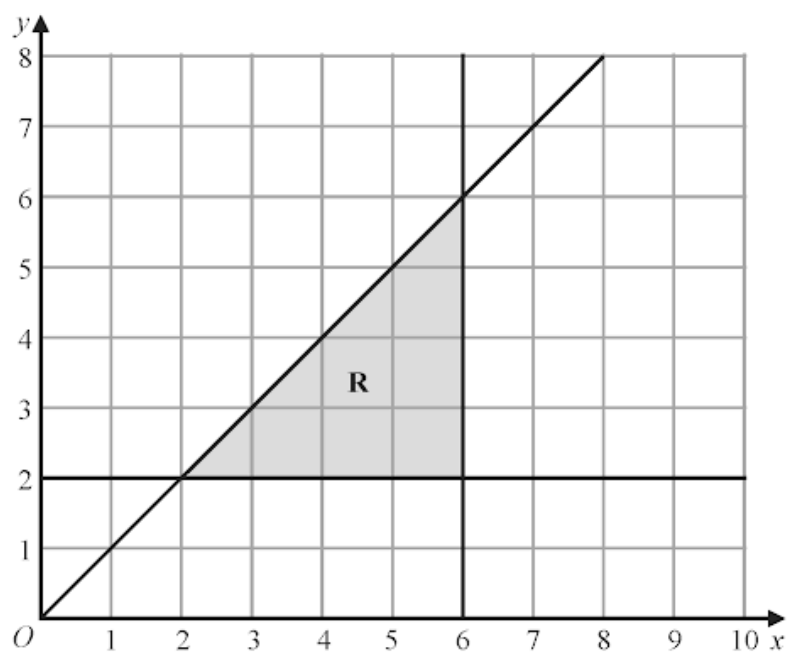
$j =$

$k =$

3 marks

(a) Solve $9 - 4x > 17$

(2)



(b) Write down the three inequalities that represent the shaded region **R**

(3)

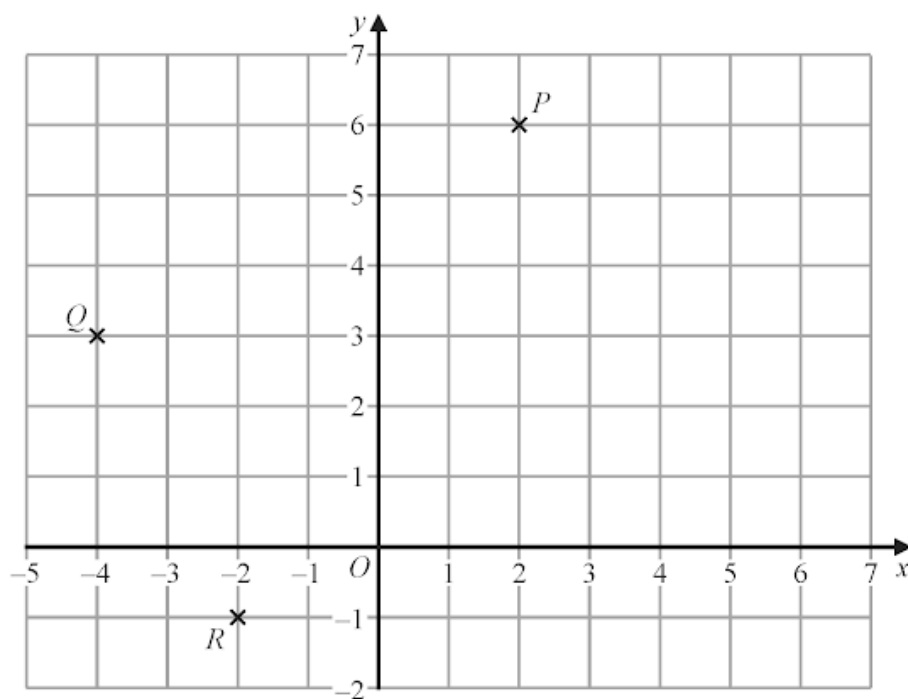
10. 4MA1/2FR_Summer_2023_Q19

Here are the first four terms of an arithmetic sequence.

38 31 24 17

Find an expression, in terms of n , for the n th term of the sequence.

.....
2 marks



(a) Write down the coordinates of the point

(i) P

(.....,)
(1)

(ii) Q

(.....,)
(1)

(b) On the grid above, mark with a cross ($\times$) the point S so that $PQRS$ is a rectangle.

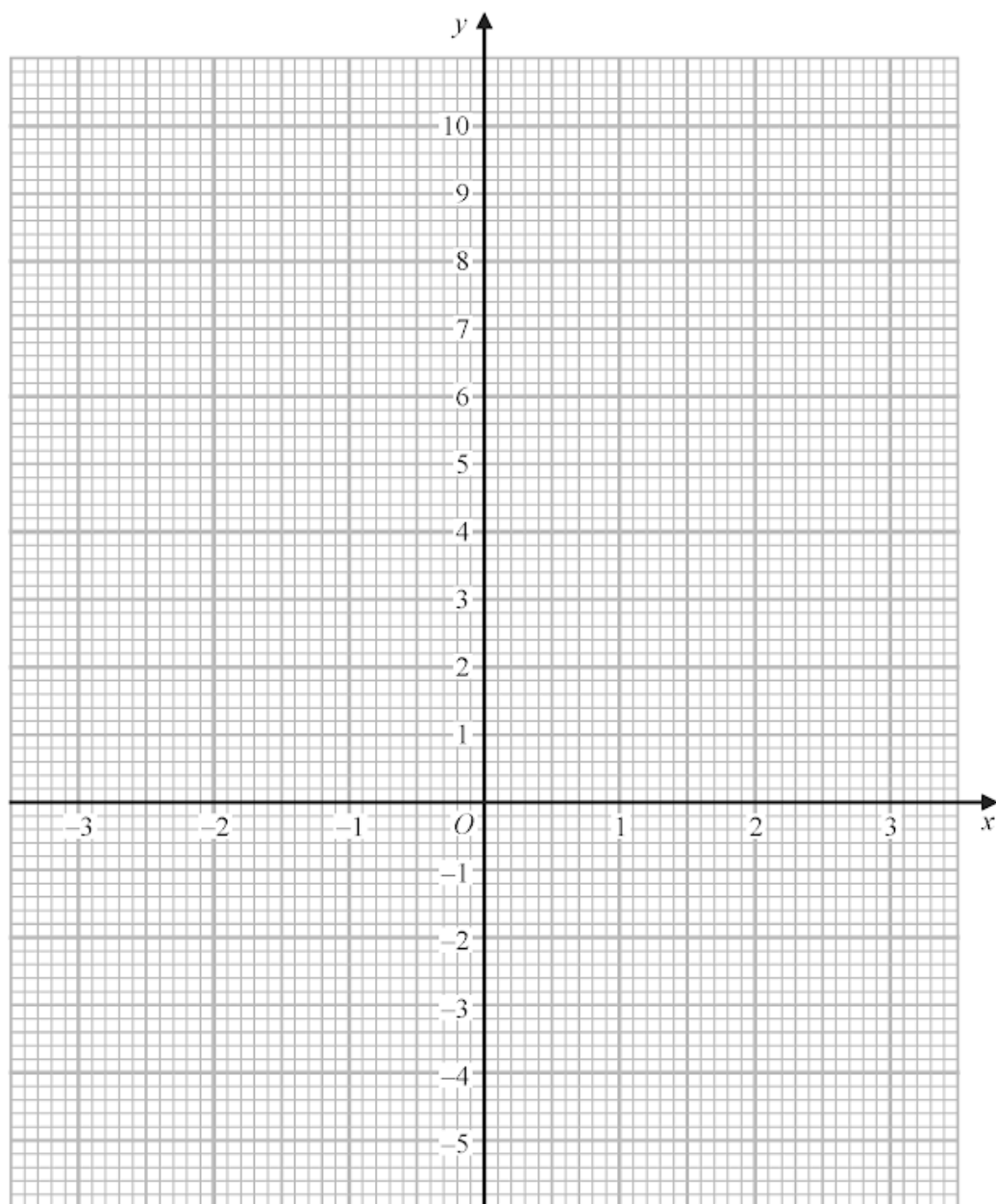
(1)

(a) Complete the table of values for $y = x^2 - x - 4$

x	-3	-2	-1	0	1	2	3
y		2			-4		

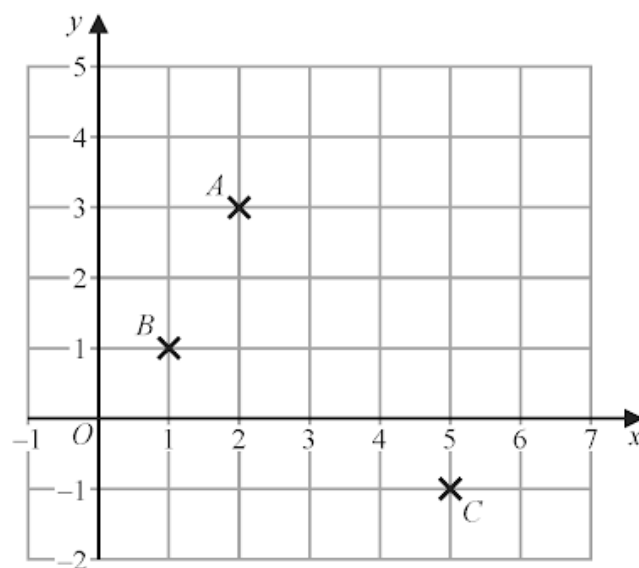
(2)

(b) On the grid below, draw the graph of $y = x^2 - x - 4$ for values of x from -3 to 3



(2)

The diagram shows three points, A , B and C , marked on a grid.



(a) Write down the coordinates of point A

(..... ,)
(1)

(b) On the grid, mark with a cross ($\times$) the point D so that $ABCD$ is a rectangle.

(1)

(c) Find the coordinates of the midpoint of AC

(..... ,)
(2)

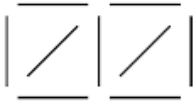
(d) On the grid, draw the line with equation $y = 4$

(1)

Here is a sequence of patterns made from sticks.



Pattern number 1



Pattern number 2



Pattern number 3

(a) In the space below, draw Pattern number 4

(1)

(b) Complete the table.

Pattern number	1	2	3	4	5
Number of sticks	5	9	13		

(1)

(c) Work out the number of sticks in Pattern number 10

(1)

Connor says that in Pattern number 25 there are 102 sticks.

(d) Explain why Connor is wrong.

(1)

- (a) Write down an equation of the straight line with gradient -3 and which passes through the point with coordinates $(0, 5)$

.....
(2)

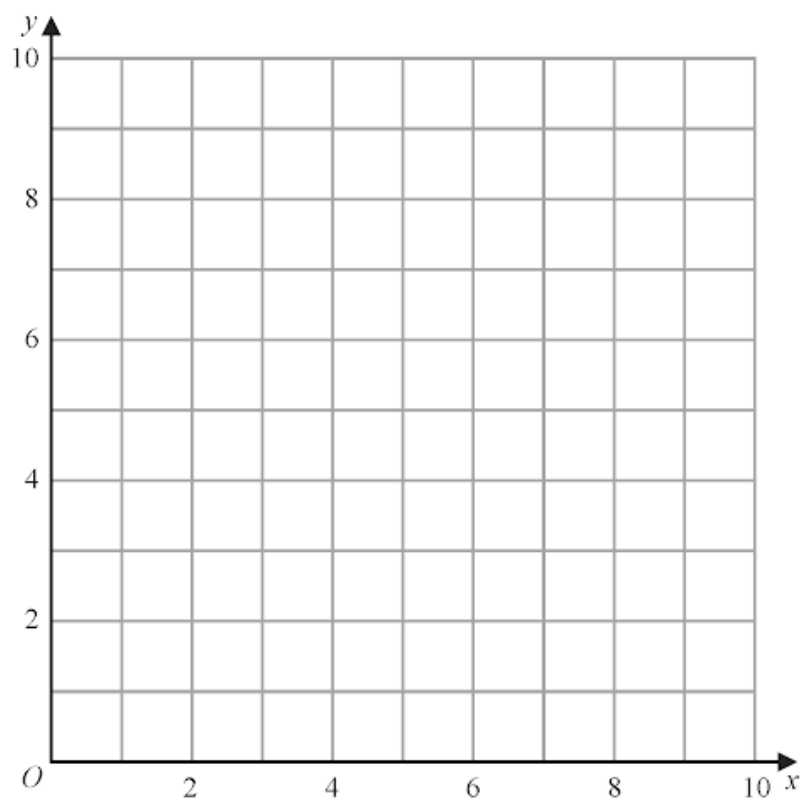
- (b) Show, by shading on the grid, the region defined by **all three** of the inequalities

$$x \leq 6$$

$$y \geq 2$$

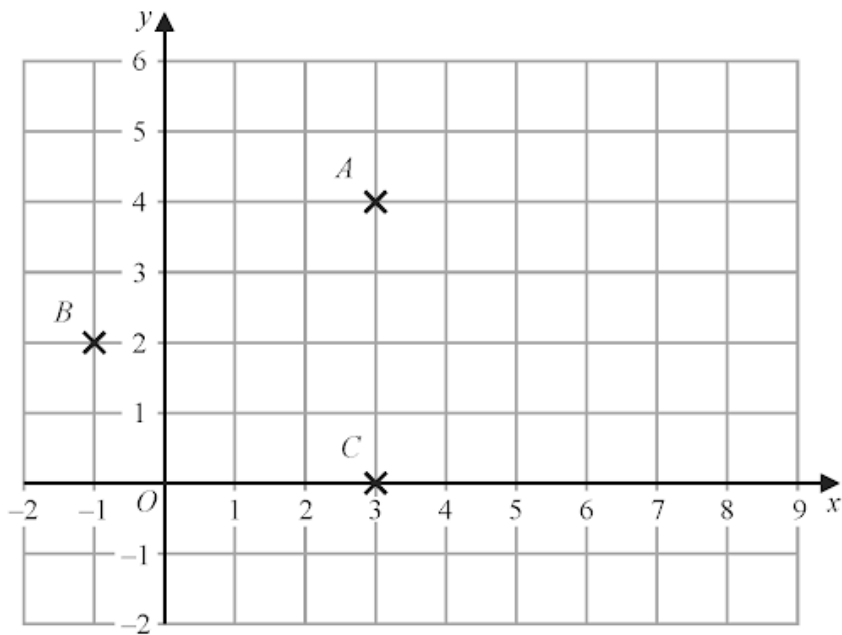
$$y \leq x + 1$$

Label the region **R**



(3)

The diagram shows three points, A , B and C , on a grid.



(a) Write down the coordinates of

(i) point A

(..... ,)

(ii) point B

(..... ,)
(2)

D is the point such that $ABCD$ is a rhombus.

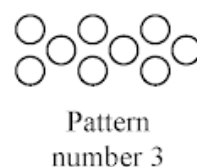
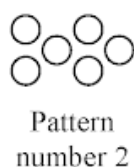
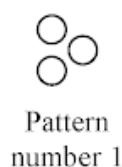
(b) On the grid, mark with a cross ($\times$) the point D
Label this point D

(1)

(c) Find the coordinates of the midpoint of AB

(..... ,)
(2)

Here is a sequence of patterns made from counters.



(a) In the space below, draw Pattern number 4

(1)

(b) Complete the table.

Pattern number	1	2	3	4	5
Number of counters	3	6	9		

(1)

(c) Work out the number of counters in Pattern number 10

(1)

Sven has exactly 70 counters.

(d) Can Sven make Pattern number 25 using his 70 counters?

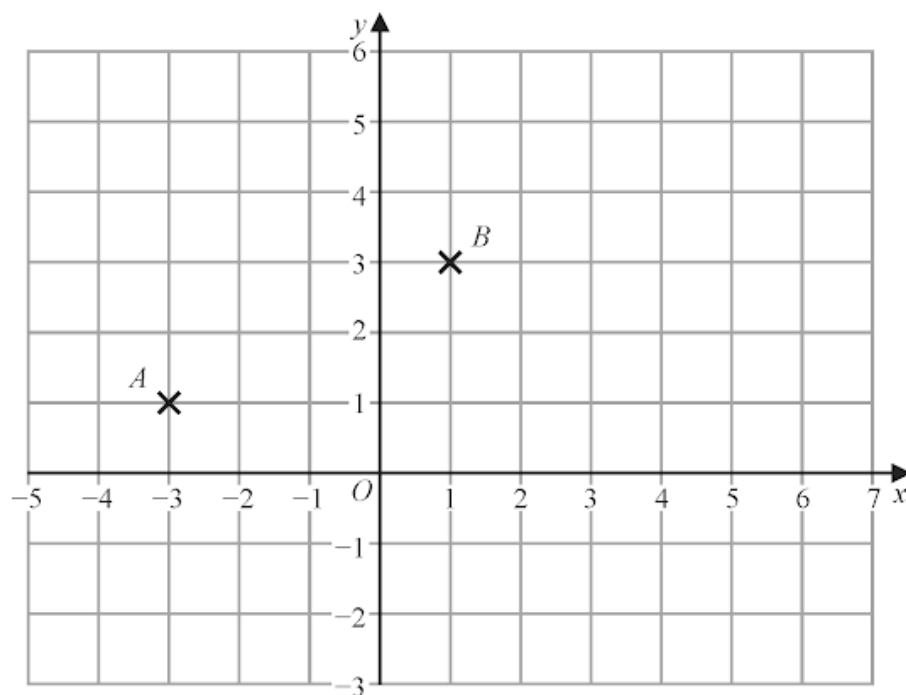
Tick the appropriate box below.

Yes	No
☐	☐

Give a reason for your answer.

(1)

The diagram shows points A and B marked on a grid of squares.



- (a) On the grid, draw the line with equation $y = -2$

(1)

M is the midpoint of AB

- (b) Find the coordinates of M

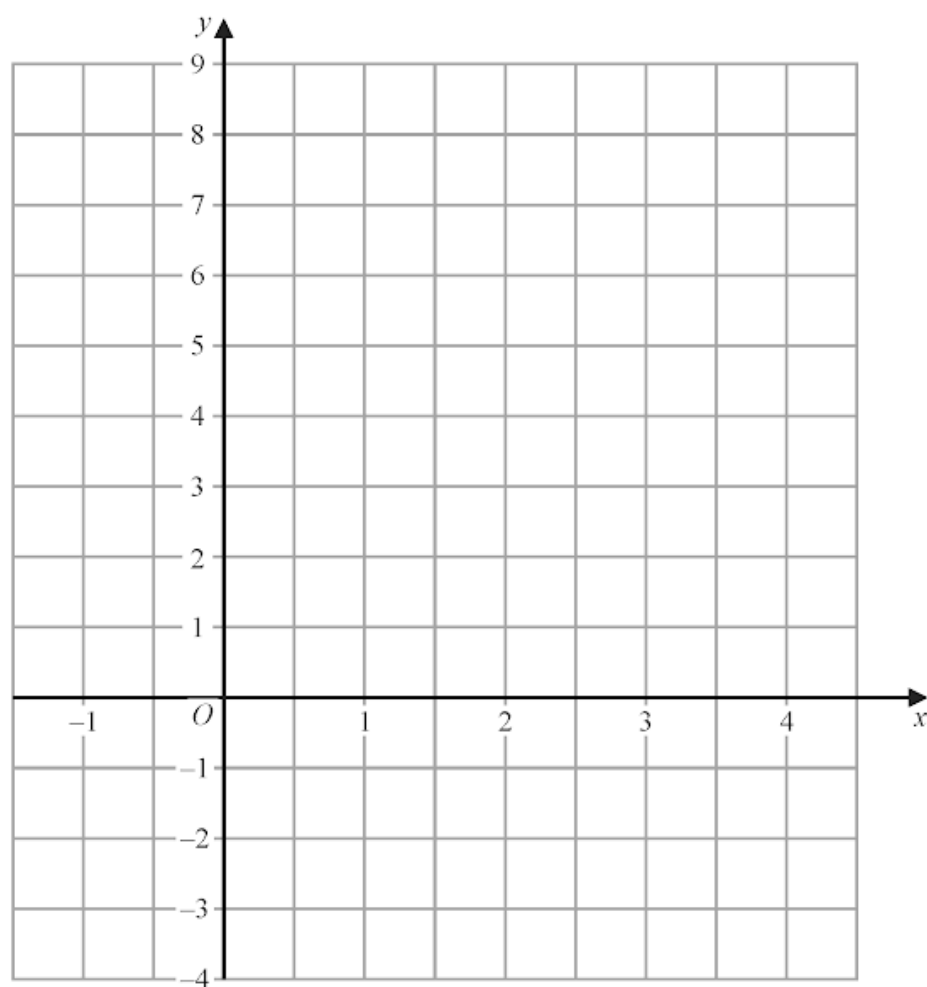
(.....,)
(2)

D is the point with coordinates $(5, d)$ where $d > 0$
The triangle ABD is an isosceles triangle.

- (c) Find the value of d

$d =$
(1)

On the grid, draw the graph of $y = 2x - 1$ for values of x from -1 to 4



3 marks

Here are the first five terms of a number sequence.

1 7 13 19 25

(a) (i) Write down the next term of the sequence.

.....
(1)

(ii) Explain how you worked out your answer.

.....
(1)

(b) Explain why 188 cannot be a number in the sequence.

.....
(1)

.....

.....

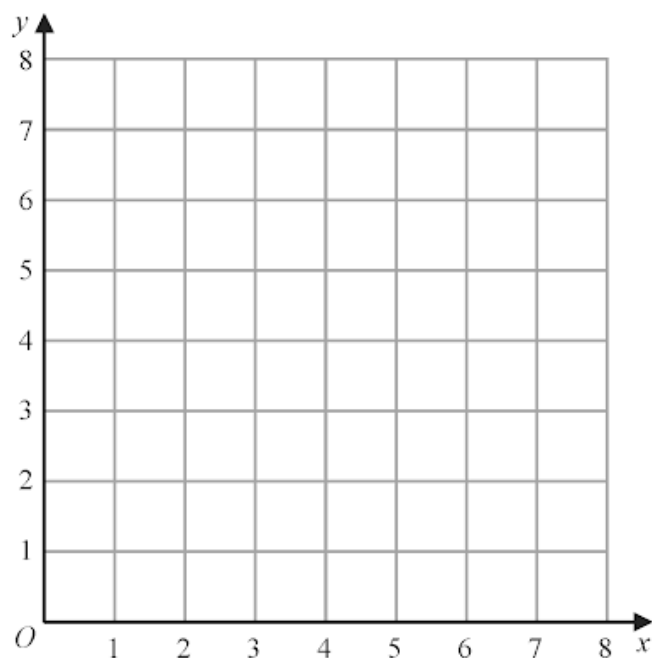
.....

.....

.....

(a) On the grid, draw and label with its equation the straight line with equation

(i) $y = 1$ (ii) $x = 2$ (iii) $x + y = 7$



(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$y \geq 1$ $x \geq 2$ $x + y \leq 7$

Label the region **R**.

(1)

Here are the first five terms of a number sequence.

7 13 19 25 31

(a) (i) Write down the next term of the sequence.

.....
(1)

(ii) Explain how you found your answer to part (a)(i)

.....
(1)

The 30th term of the sequence is 181

(b) Work out the 28th term of the sequence.

.....
(1)

Brian says that 96 is a number in the sequence.

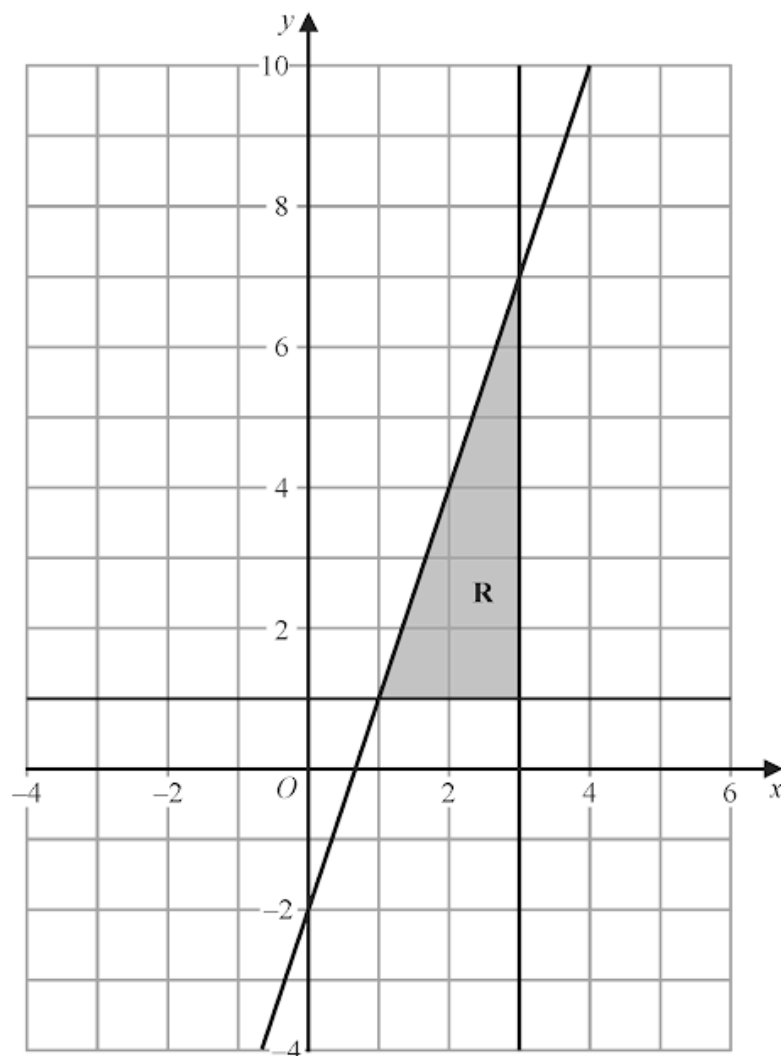
Brian is wrong.

(c) Explain why.

.....
.....
(1)
.....
.....
.....
.....
.....

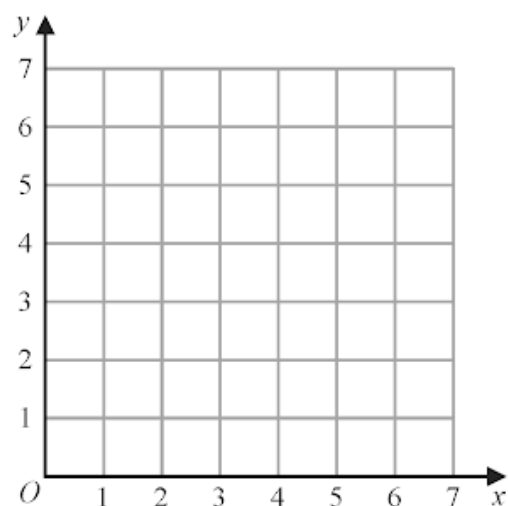
The shaded region **R**, shown in the diagram below, is bounded by the straight line with equation $y = 3x - 2$ and by two other straight lines.

Write down the three inequalities that define region **R**.



.....
.....
.....

3 marks



(a) On the grid, draw and **label** the straight line with equation

(i) $x = 1.5$

(ii) $y = x$

(iii) $x + y = 6$

(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$$x \geq 1.5$$

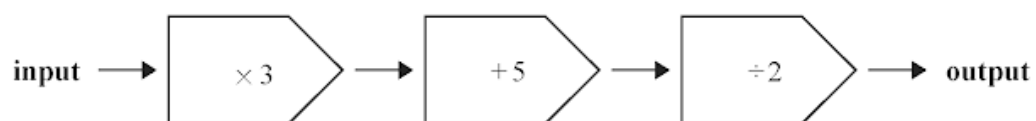
$$y \geq x$$

$$x + y \leq 6$$

Label the region **R**.

(1)

Here is a number machine.



(a) Work out the output when the input is 7

.....
(1)

(b) Work out the input when the output is 160

.....
(2)

When the input is n , the output is P .

(c) Find a formula for P in terms of n .

.....
(2)

.....

.....

.....

.....

.....

26. 4MA1/2F_Summer_2021_Q4

27. 4MA1/2FR_Winter_2020_Q25

28. 4MA1/2FR_Winter_2020_Q9

29. 4MA1/2FR_Winter_2020_Q7

30. 4MA1/2F_Winter_2020_Q8

31. 4MA1/2F_Winter_2020_Q7

32. 4MA1/2FR_Summer_2020_Q24

33. 4MA1/2FR_Summer_2020_Q3

34. 4MA1/2FR_Summer_2020_Q7

35. 4MA1/2F_Summer_2020_Q6

36. 4MA1/2FR_Winter_2019_Q17

37. 4MA1/2FR_Winter_2019_Q24

38. 4MA1/2FR_Summer_2019_Q26

39. 4MA1/2FR_Summer_2019_Q19

40. 4MA1/2FR_Summer_2019_Q15

41. 4MA1/2FR_Summer_2019_Q7

42. 4MA1/2F_Summer_2019_Q12

43. 4MA1/2F_Summer_2019_Q6

44. 4MA1/2FR_Summer_2018_Q14

45. 4MA1/2F_Summer_2018_Q12

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